

European Telescope Discovers ‘Einstein Ring’ in Nearby Galaxy

Scientists say Europe’s Euclid space telescope has discovered a bright ring of light circling a nearby galaxy. The observed light is known as an Einstein ring.

Researchers have estimated the ring encircles a galaxy 590 million light years away from Earth. A light year is the distance light travels in a year – about 9.5 trillion kilometers.

Astronomers have long known about the galaxy where the ring was discovered. So, they were surprised that the bright ring had not been identified before.

It was discovered in a well-studied galaxy called NGC 6505. It is in the constellation – a group of stars – named Draco.

Einstein rings are rare. They form when light from a more distant galaxy bends around a closer galaxy. The nearer galaxy appears to have a circle of light, or a halo, around it.

This artist's impression depicts the European Space Agency's Euclid spacecraft, which is carrying out a mission to explore the "Dark Side" of the universe. (Image Credit: ESA)

The Einstein ring gets its name from physicist Albert Einstein. Einstein predicted that light would bend around extremely massive objects in space. This is called “gravitational lensing.”

The American space agency NASA explains that as light passes through a gravitational lens, “it may take different paths, producing multiple images of the same object.” In this way, gravity itself acts as a lens, magnifying and distorting space and time in a way that is similar to an optical lens like those in eyeglasses or contact lenses.

Gravitational lenses permit telescopes like Euclid to observe more distant and less-bright objects. In the latest project, researchers from Germany’s Max Planck Institute for Astrophysics used data from Euclid to create a computer model to discover the Einstein ring.

The Euclid space telescope is operated by the European Space Agency (ESA). In 2023, it launched on a six-year mission that ESA officials have said aims to study the mysteries of dark matter and dark energy across the universe.

This photo provided by the European Space Agency shows a close-up view of the center of the NGC 6505 galaxy, with the bright Einstein ring aligned with it, captured by European Space Agency's Euclid space telescope. (European Space Agency via AP)

Astronomers at the Max Planck Institute are leading the research. In a statement, the organization said the first sign that the Einstein ring existed came during testing right after Euclid was deployed. A team member, Bruno Altieri, was looking over early data collected by Euclid.

“Even from that first observation, I could see it,” Altieri said in a statement. “But after Euclid made more observations of the area, we could see a perfect Einstein ring. For me, with a lifelong interest in gravitational lensing, that was amazing.”

The scientists said the more distant galaxy involved in the method is about 4.4 billion light years away from Earth. They noted that the more distant galaxy had never been observed before and does not have a name.

Altieri’s team recently reported the findings in a study in the publication *Astronomy and Astrophysics*.

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Conor O’Riordan is another team member from the Max Planck Institute and was the lead writer of the study. He said, “All strong lenses are special, because they're so rare, and they're incredibly useful scientifically. This one is particularly special, because it’s so close to Earth and the alignment makes it very beautiful.”

O’Riordan added that the researchers used the latest gravitational lensing methods developed at the institute. This permitted them to model the light of the ring. “We even had to look at some of the raw data to better understand how the detector works.”

O’Riordan said the team’s modeling operations were just the first step in studying the newly discovered Einstein ring. He said the group plans to use the ring to also study “dark matter substructures” within the lensing galaxy.

O’Riordan predicted, “Euclid is going to revolutionize the field, with all this data we've never had before.”

I’m Bryan Lynn.

Bryan Lynn wrote this story for VOA Learning English, based on reports from The Associated Press, the Max Planck Institute, the European Space Agency and NASA.

Vocabulary

galaxy - n. ดาราจักร, กาแล็กซี (กลุ่มดาวขนาดใหญ่ที่รวมตัวกันในจักรวาล)

bend - v. โค้งงอ, ดัดให้โค้ง

distort - v. บิดเบือน, ทำให้ผิดรูป/ผิดเพี้ยน

lens - n. เลนส์ (ชิ้นส่วนแก้วโค้งที่ใช้ในกล้อง แว่นตา และอุปกรณ์วิทยาศาสตร์)

magnify - v. ขยาย (ทำให้วัตถุใหญ่ขึ้นผ่านอุปกรณ์พิเศษ)

perfect - adj. สมบูรณ์แบบ, ไม่มีข้อผิดพลาด, ถูกต้องครบถ้วน

align - v. จัดเรียงให้ตรงกัน, ทำให้เป็นแนวเดียวกัน

raw - n. สิ่งที่ยังไม่ผ่านการเตรียมหรือแปรรูป, วัตถุดิบ (ตามที่ระบุในคำนิยามแม้จะใช้เป็นคำคุณศัพท์ในบริบทของบทความ)

detect - v. ตรวจจับ, ค้นพบ (ระบุบางสิ่งโดยเฉพาะสิ่งที่มองเห็น/ได้ยิน/ได้กลิ่นยาก)

Reading Comprehension Quiz

1. What has Europe's Euclid space telescope recently discovered?

- A) A new galaxy
- B) A black hole
- C) A bright ring of light
- D) A distant planet

2. How far away is the galaxy that the discovered Einstein ring encircles?

- A) 9.5 trillion kilometers
- B) 590 million light years
- C) One light year
- D) Undetermined

3. What was surprising to astronomers about the bright ring?

- A) Its unusual shape
- B) Its extreme brightness
- C) That it had not been identified earlier
- D) Its rapid movement

4. In which specific galaxy and constellation was the Einstein ring discovered?

- A) NGC 6505, in the constellation Orion
- B) A galaxy 590 million light years away, in Draco
- C) NGC 6505, in the constellation Draco
- D) An unnamed galaxy, in the constellation Hydra

5. According to the article, how do Einstein rings form?

- A) When two galaxies collide

- B) From the reflection of light off a black hole
- C) When light from a distant galaxy bends around a closer one
- D) Due to supernovas occurring in a cluster of stars

6. From whom does the Einstein ring get its name, and what related phenomenon did he predict?

- A) Isaac Newton; the law of gravity
- B) Albert Einstein; gravitational lensing
- C) Galileo Galilei; the phases of Venus
- D) Stephen Hawking; black hole radiation

7. How does NASA explain the effect of a gravitational lens on light?

- A) It absorbs light, making objects invisible.
- B) It causes light to travel in a straight line, making objects clearer.
- C) It makes light take different paths, producing multiple images and distorting space and time.
- D) It reflects light, creating illusions of objects that aren't there.

8. What is one key advantage that gravitational lenses offer to telescopes like Euclid?

- A) They make objects appear closer than they are.
- B) They allow observation of more distant and less-bright objects.
- C) They reduce the need for powerful telescope optics.
- D) They block out harmful radiation from space.

9. Who operates the Euclid space telescope, and what is the primary goal of its six-year mission?

- A) NASA; to find extraterrestrial life.
- B) Max Planck Institute; to discover new galaxies.
- C) European Space Agency (ESA); to study dark matter and dark energy.
- D) The Associated Press; to document space discoveries.

10. Who is leading the research on the Einstein ring discovery, and when did the first sign of its existence emerge?

- A) NASA; during the telescope's launch.
- B) European Space Agency; before Euclid was deployed.
- C) Max Planck Institute; during testing right after Euclid was deployed.
- D) Bruno Altieri's personal project; after years of data analysis.

11. What was Bruno Altieri's initial reaction to seeing the early data from Euclid?

- A) He was confused by the unexpected image.
- B) He immediately recognized a 'perfect Einstein ring.'
- C) He saw a sign of the ring and was amazed after more observations confirmed it.
- D) He dismissed it as a data error.

12. How far away is the more distant galaxy involved in the gravitational lensing, and what is notable about its previous observation?

- A) 590 million light years; it was well-studied.
- B) 4.4 billion light years; it had never been observed before.
- C) 9.5 trillion kilometers; it was discovered by accident.
- D) Unspecified; it is a known galaxy in Draco.

13. In which publication did Altieri's team report their findings?

- A) VOA Learning English
- B) The Associated Press
- C) Astronomy and Astrophysics
- D) Max Planck Institute Journal

14. According to Conor O'Riordan, what makes this newly discovered Einstein ring 'particularly special'?

- A) It is the largest Einstein ring ever found.
- B) Its advanced modeling operations.
- C) Its proximity to Earth and beautiful alignment.
- D) It disproves previous theories about gravitational lensing.

15. What future research plan does O'Riordan mention regarding the newly discovered Einstein ring?

- A) To use it to find new constellations.
- B) To study 'dark matter substructures' within the lensing galaxy.
- C) To determine its exact age and origin.
- D) To develop new telescope technologies.